Homework 3

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Loading of Health data

```
library(tidyverse)
options(scipen = 999)
statme <- read_table("statme.txt")

# Ensure X is properly defined (5 columns of statme)
#Manipulating the data set to create a constant variable for column fill #with 1, then combaining it with c
X<- statme %>%
  mutate(constant = rep(1, 53)) %>%
  select(6,2:5) %>%
  as.matrix()

# Ensure Y is properly defined (first 4 columns of statme)
Y <- as.matrix(statme[, 1])
```

1.

$$X'X =$$

```
# Compute X'X
XtX <- t(X) %*% X

# Print the result
print(XtX)
```

```
##           constant        X2          X3        X4        X5
## constant      53.0    6153.0     31259.0    500.10    5864.0
## X2          6153.0  788969.0   3822716.0  58976.80  678924.0
## X3         31259.0 3822716.0  24189339.0 295466.80 3610827.0
## X4           500.1   58976.8    295466.8   4779.01   55671.5
## X5          5864.0  678924.0   3610827.0  55671.50  764550.0
```

2.

$$(X'X)^{-1} =$$

```
XtX_inv <- solve(XtX)
print(XtX_inv)
```

```
##                constant               X2               X3            X4
## constant   1.5916252455  0.0007434977083 -0.0001043396745 -0.16554210565
```

```
## X2        0.0007434977  0.0000187498773 -0.0000006589822 -0.00029202986
## X3       -0.0001043397 -0.0000006589822  0.0000002032566  0.00001006668
## X4       -0.1655421057 -0.0002920298615  0.0000100666794  0.02145617803
## X5       -0.0003208934  0.0000020241955 -0.0000003075074 -0.00008088635
##                     X5
## constant -0.0003208934272
## X2        0.0000020241955
## X3       -0.0000003075074
## X4       -0.0000808863494
## X5        0.0000093137985
```

3.

$$X'Y =$$

```r
XtY <- t(X) %*% Y
print(XtY)
```

```
##                 X1
## constant    493.20
## X2        57636.90
## X3       294065.70
## X4         4637.77
## X5        53436.10
```

4.

$$b = (X'X)^{-1}X'Y$$

```r
b <- XtX_inv %*% XtY
colnames(b) <- c("Estimates")
rownames(b) <- c("Intercept", "Doctor Availability", "Hospital Availability", "Income", "Population Density

print(b)  # Regression coefficients
```

```
##                           Estimates
## Intercept              12.2662551669
## Doctor Availability     0.0073916150
## Hospital Availability   0.0005837157
## Income                 -0.3302302366
## Population Density     -0.0094628849
```

5. Linear regression model:

$$\text{death rate} = \beta_0 + \beta_1\text{doctor availability} + \beta_2\text{hospital availability} + \beta_3\text{income} + \beta_4\text{population density}$$

$$\text{death rate} = 12.2662552 + 0.0073916(\text{doctor availability}) + 0.0005837(\text{hospital availability}) - 0.3302302(\text{income}) - 0.0094629(\text{popula}$$